

# MVC Composite Dielectric

Meta-Void Composite Laminate | Low-Loss RF/High-Speed PCB Substrate | V5.0

MVC-DS-001-V5.0

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License: AGPL-3.0 + CERN-OHL-S

| DK @ 10 GHz | DF @ 10 GHz | DENSITY                   |
|-------------|-------------|---------------------------|
| 2.98        | 0.0034      | 1.68<br>g/cm <sup>3</sup> |

| KEY ELECTRICAL PROPERTIES |               |
|---------------------------|---------------|
| Dk @ 1 GHz                | 3.10 ± 0.05   |
| Dk @ 5 GHz                | 3.05 ± 0.05   |
| Dk @ 10 GHz               | 2.98 ± 0.05   |
| Dk @ 77 GHz               | 2.88 ± 0.08   |
| Df @ 1 GHz                | 0.0028        |
| Df @ 10 GHz               | 0.0034        |
| Df @ 77 GHz               | 0.0044        |
| Surf. resist.             | >=1E9 MOhm/sq |
| TDK                       | -8 ppm/C      |

| THERMAL PROPERTIES |            |
|--------------------|------------|
| Glass trans. Tg    | 185 ± 5 C  |
| Therm. cond. Z     | 0.38 W/m·K |
| Therm. cond. XY    | 0.45 W/m·K |
| CTE X-Y            | 14 ppm/C   |
| CTE Z (below Tg)   | 42 ppm/C   |
| Spec. heat Cp      | 1.05 J/g·K |

| MECHANICAL PROPERTIES |                           |
|-----------------------|---------------------------|
| Flexural modulus      | 18.5 GPa                  |
| Flexural strength     | 320 MPa                   |
| Tensile modulus       | 16.8 GPa                  |
| Compressive str.      | 410 MPa                   |
| Fracture tough. KIC   | 0.85 MPa m <sup>0.5</sup> |
| Poisson ratio         | 0.18                      |
| Hardness (Rockwell M) | 88 HRM                    |
| Impact (Izod)         | 28 kJ/m <sup>2</sup>      |
| Density               | 1.68 g/cm <sup>3</sup>    |
| Peel str. (1oz Cu)    | >=1.4 N/mm                |
| Water abs. (24h)      | 0.08 %                    |
| HAD                   | <5 µm                     |

| PROCESSING PARAMETERS |                         |
|-----------------------|-------------------------|
| Cure temp.            | 175–185 C               |
| Cure pressure         | 2.5–3.5 MPa             |
| Cure time             | 60 min                  |
| UV laser (355nm)      | 20–35 µJ / 3–5 shots    |
| CO2 laser             | 4–6 mJ / 8–12 shots     |
| Bond film             | MVC-BF-40 (40 µm)       |
| Recommended Cu        | RTF / VLP-2, Rz<=1.0 µm |
| Storage               | 15–25 C, <50% RH        |
| Shelf life            | 12 months sealed        |

| MATERIAL COMPOSITION             |              |
|----------------------------------|--------------|
| Fluoropolymer-mod. epoxy matrix  | 52 ± 2 vol%  |
| Hollow silica microspheres (HSM) | 35 ± 3 vol%  |
| E-glass woven (2116)             | 10 ± 1 vol%  |
| Aminosilane coupling agent       | 0.5–1.5 vol% |
| Phos. flame retardant            | 2–3 vol%     |

| APPLICATION NOTES                                      |  |
|--------------------------------------------------------|--|
| • RF / mmWave front-end modules (sub-6 GHz to 140 GHz) |  |
| • High-speed SerDes substrates (56+ Gbps PAM4)         |  |
| • Compute package interposers and carrier boards       |  |
| • MEMS integration platforms (low CTE mismatch)        |  |
| • Power delivery networks requiring stable Dk          |  |
| • Automotive radar modules (77/79 GHz qualified)       |  |
| • Drop-in for Rogers RO4003C / Isola I-Speed           |  |

| SAFETY & HANDLING |                                 |
|-------------------|---------------------------------|
| Flammability:     | UL 94 V-0 (halogen-free)        |
| Machining dust:   | Respirable silica — P100 resp.  |
| Skin/eye:         | Nitrile gloves + safety glasses |
| Waste:            | Non-hazardous (non-leaching)    |
| Incompatible:     | Conc. H2SO4, fuming HNO3, DMF   |
| ESD (bare):       | Not required                    |

| COMPLIANCE & QUALIFICATION                     |  |
|------------------------------------------------|--|
| • IPC-4101E   IEC 62562   UL 94 V-0            |  |
| • IPC-TM-650 (Dk, Df, peel, water abs., IST)   |  |
| • ASTM D792, D638, D695, E399, E1461           |  |
| • CAF resistance: <1 µA @ 500h (IPC 2.6.25)    |  |
| • Thermal shock: No delamination (IPC 2.6.7.2) |  |